

Fat Soluble Vitamin

Vitamin D

SUNLIGHT
The body itself makes
vitamin D when it is
exposed to the sun

10 Best Vitamin D Rich Foods



Fortified Cereals



Egg Yolk



Mushroom



Ricotta Cheese



Fatty Fish



COD Liver Oil



Cavier (Fish eggs)

Vitamin D:
‘The sunshine vitamin’

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Objectives: DLA: Vitamin D

- Review DLA before the lecture: Calcium, phosphate and vitamin D

SOM.MK.III.BPM1.2.MSK.1.BCHM.0200	Describe vitamin D formation in the skin and its conversion to calcitriol (in liver and kidney) indicating factors affecting this process.
SOM.MK.III.BPM1.2.MSK.1.BCHM.0201	Describe the functions of 1,25 dihydroxy-D (calcitriol) related to calcium and phosphate metabolism
SOM.MK.III.BPM1.2.MSK.1.BCHM.0202	Indicate vitamins that act as steroid hormones (calcitriol)
SOM.MK.III.BPM1.2.MSK.1.BCHM.0203	Indicate clinical features and manifestations of hypervitaminosis D
SOM.MK.III.BPM1.2.MSK.1.BCHM.0204	Describe the basis of clinical manifestation in rickets and osteomalacia. Identify risk factors for development of these disorders.

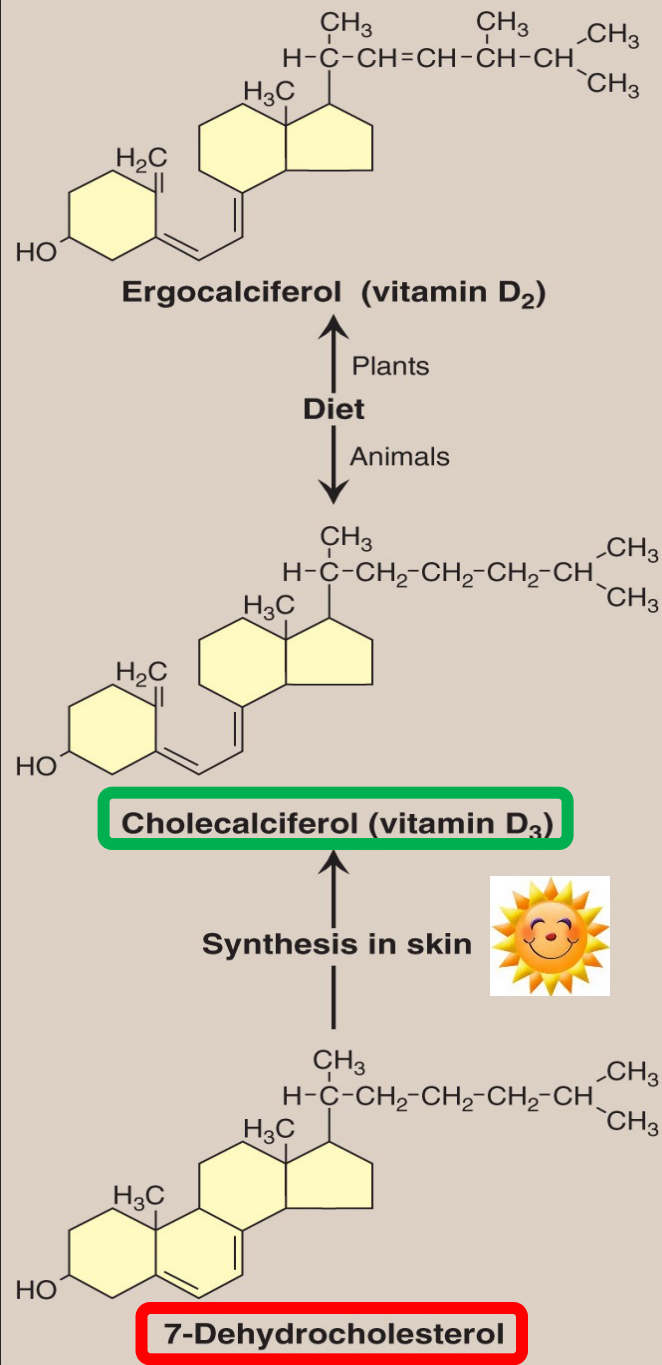
Lippincott's 9th edition:

[Micronutrients: Vitamins](#) | [Lippincott Illustrated Reviews: Biochemistry, 9e](#) | [Premium Basic Sciences](#) | [Health Library](#);

Section : **Vitamin D**

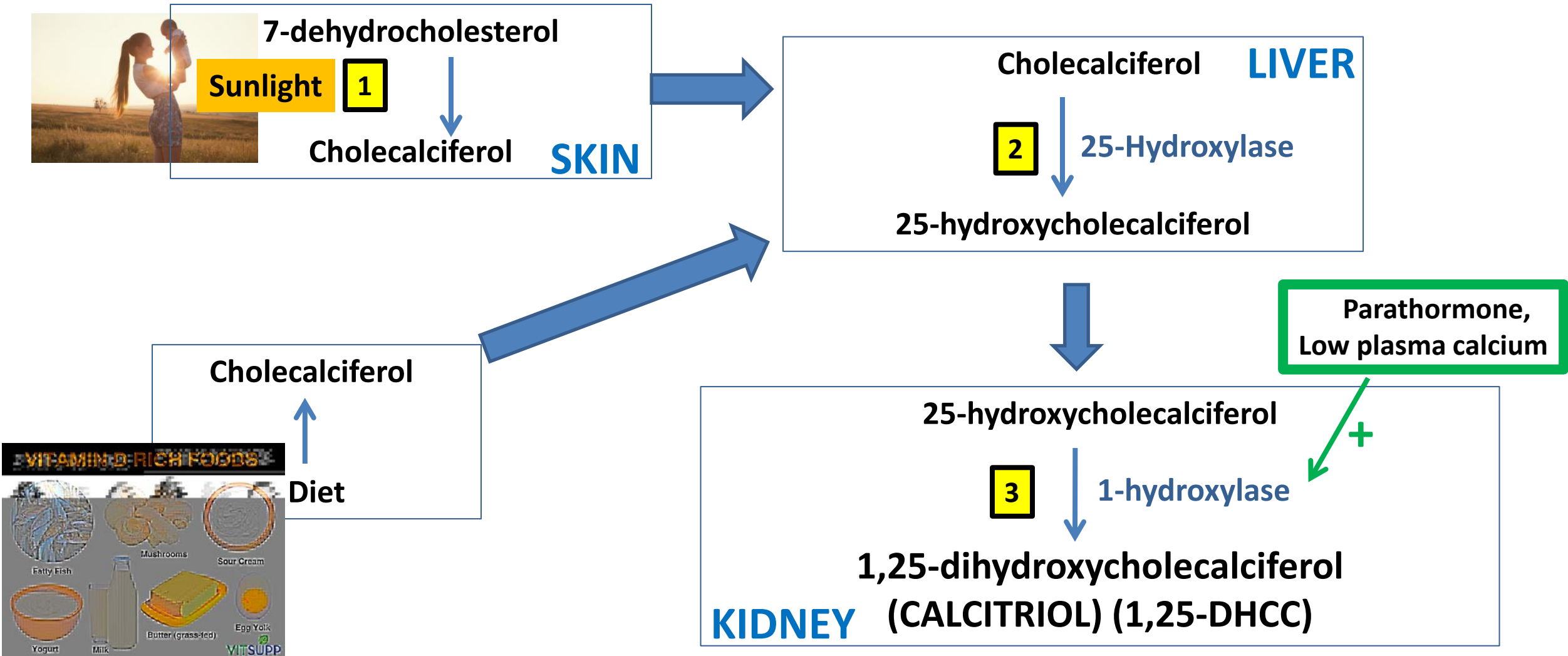
<https://emedicine.medscape.com/article/128762-overview#a6>

VITAMIN D (Calciferol)



- Group of sterols that have a hormone like function
- **Ergocalciferol (vitamin D₂)** – found in plants
- **Cholecalciferol (vitamin D₃)** – Found in animal sources
- The precursor for **cholecalciferol (vitamin D₃)** synthesis in the skin is **7-dehydrocholesterol** (intermediate in cholesterol synthesis)
- Adequate exposure to sunlight can prevent vitamin D deficiency

Formation of calcitriol (1,25 dihydroxy cholecalciferol) (active form of vitamin D)



Factors affecting the formation of vitamin D in the skin

- Increased melanin pigment reduces its formation
- Time of exposure and latitude
- Amount of exposed skin/ sunscreen reduces formation
- Winter months and latitude



Inactivation of 1,25-DHCC: Action of 24-Hydroxylase

1,25-Dihydroxycholecalciferol (Calcitriol)



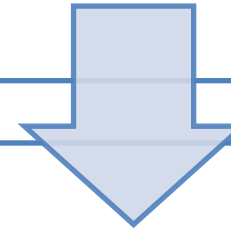
24-Hydroxylase

1,24,25-hydroxycholecalciferol (Inactive)

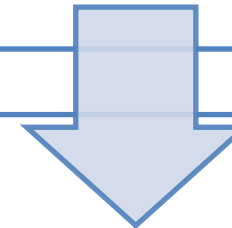
Mechanism of action of 1,25 dihydroxycholecalciferol (calcitriol)

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1,25 Dihydroxycholecalciferol (Calcitriol) binds to intracellular receptor proteins in target cells (**intestinal mucosal cells**)



1,25-DHCC receptor complex interacts with **DNA (hormone response element) in the nucleus** of target cells (**intestine**)



Can either selectively stimulate gene expression or repress gene expression – Increases synthesis of **calbindins** (**similar to mechanism of action of steroid hormones, thyroid hormone, retinoic acid**)

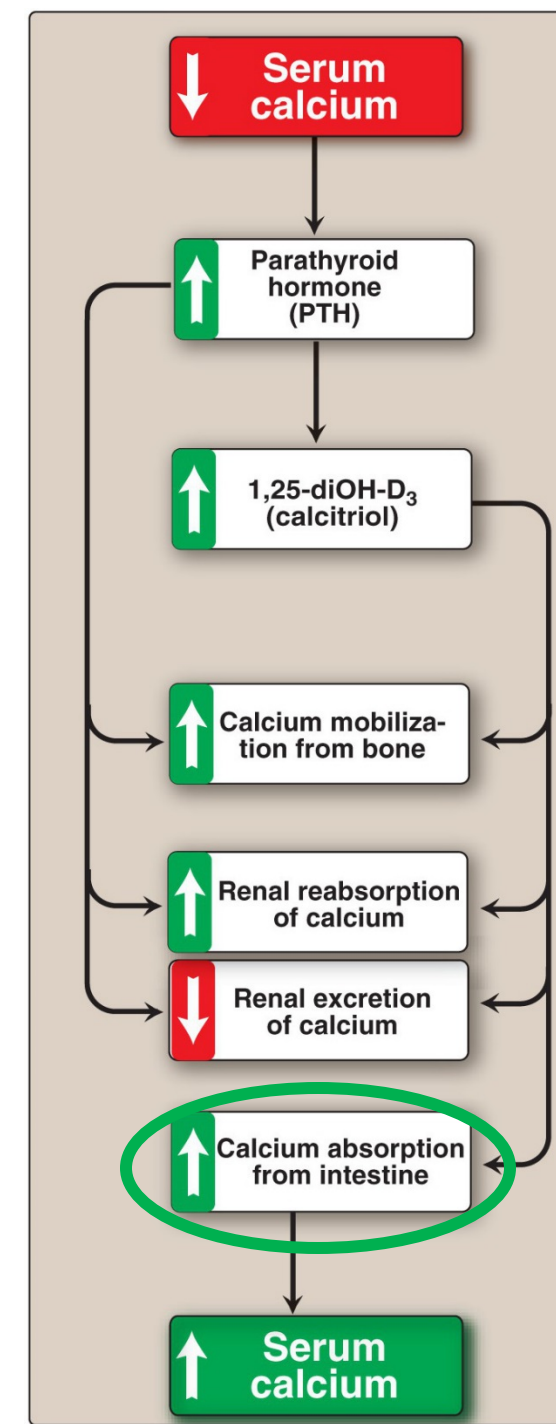
Vitamin D: Serum calcium

4

- On the **intestine** – stimulates intestinal absorption of calcium and phosphate by increased synthesis of a specific calcium binding protein (calbindin)
- On the **bone** – stimulates the mobilization of calcium and phosphate from the bone in the presence of parathormone (when the serum calcium level is low)
- On the **kidneys** – inhibits calcium excretion by stimulating parathyroid dependent calcium reabsorption

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Increases Serum Calcium



Vitamin D: Serum phosphate

- Vitamin D **increases serum phosphate** by increasing its absorption from the intestine
- In the presence of vitamin D, serum calcium and phosphate are elevated – Increases the tendency to form bone mineral (Vitamin D facilitates mineralization, when serum calcium and phosphate are adequate)
- In vitamin D deficiency, reduced absorption of calcium reduces serum calcium level – stimulates secretion of PTH, which in turn causes demineralization of bone. Remember, PTH also increases excretion of phosphate in kidney
- **Patients with vitamin D deficiency have low serum calcium and low serum phosphate levels and decreased bone mineralization**

Vitamin D deficiency

- Risk factors:
 - Inadequate exposure to sunlight (common during winter; in elderly)
 - Nutritional deficiency – decreased intake/ fat malabsorption/ exclusively breast fed infants
 - Chronic renal disease, chronic liver disease results in decreased hydroxylation of vitamin D

Rickets: (Vitamin D deficiency in children)

- Decreased calcium absorption from the diet → Decreased serum calcium → ↑Parathyroid hormone release → ↑demineralization of bone**
- Demineralization of bone – soft pliable bones
- Characteristic **bow-leg deformity**
- Overgrowth at costochondral junction – **rachitic rosary**
- **Pigeon chest deformity**
- Frontal bossing, delayed closure of fontanelle
- Delayed teething in infants



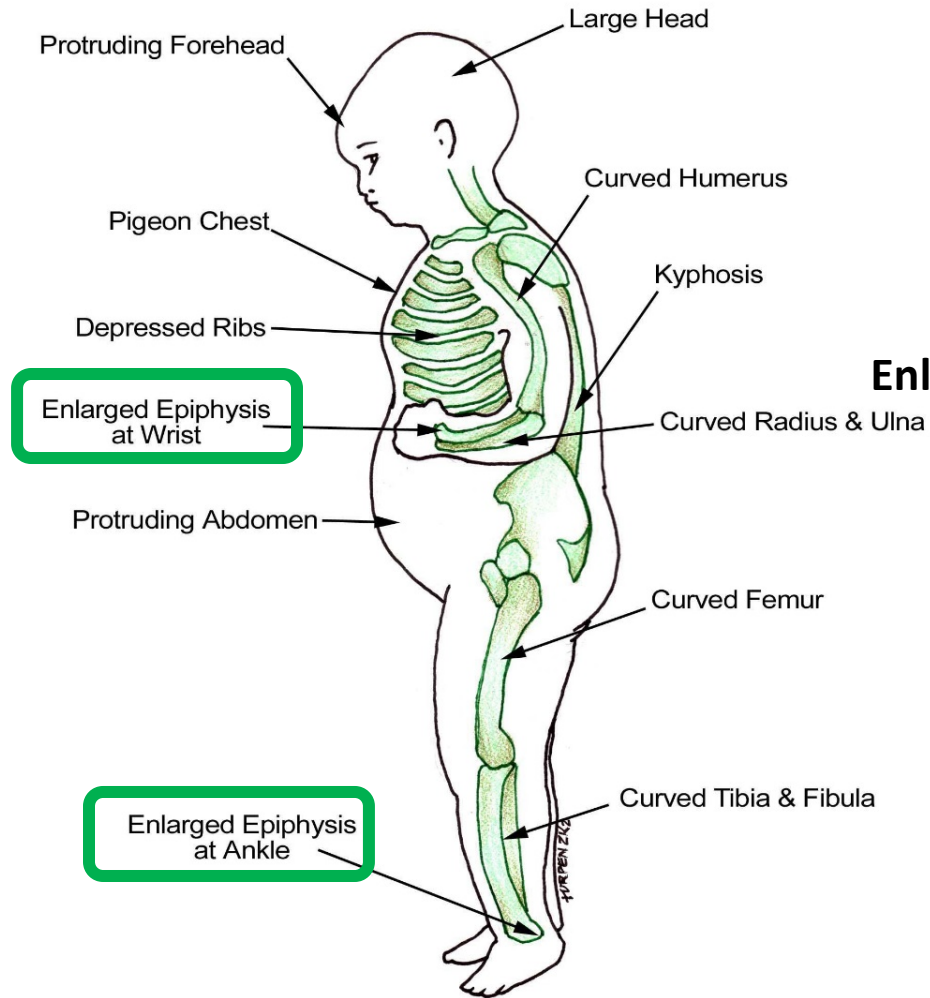
Examination of fontanelle



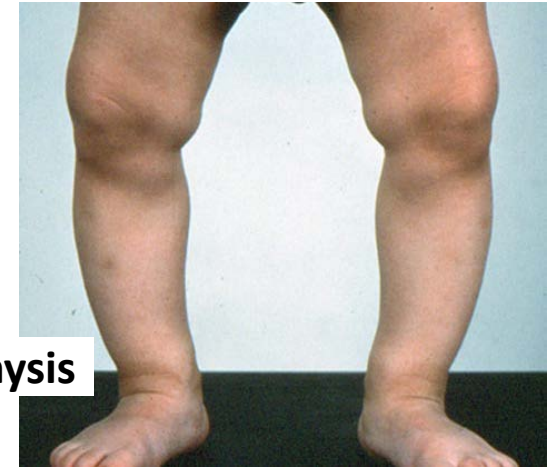
Frontal bossing



Pigeon chest deformity



Enlarged epiphysis



Bowing of legs



Bowing of legs



Rachitic rosary

Radiologic and lab findings



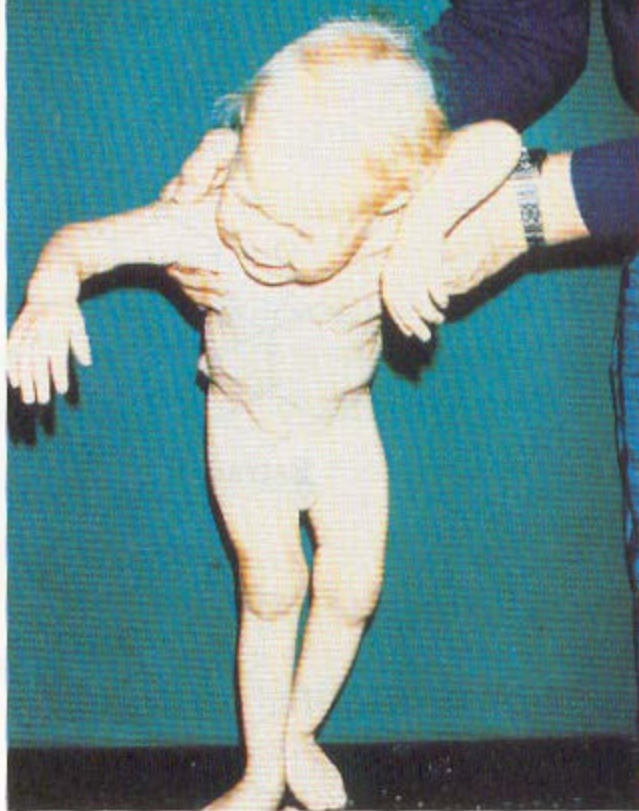
Lab findings

- Elevated **Serum alkaline phosphatase (ALP)**
- **Low serum calcium levels**
- **Low serum phosphate levels**

(A) **Rickets.** Anteroposterior radiograph of the wrist and hand in a 3-year-old child with nutritional rickets. The child had been put on a strict diet without dairy products. Note the widening, cupping, and fraying of the distal radius (arrowhead) and ulna metaphyses with an associated increase in the thickness of the growth plate (arrow). These changes are the consequence of disordered endochondral growth.

(B) **Normal.** Radiograph of the hand of a healthy 3-year-old child, without rickets.

Vitamin D and calcium supplementation



Child with rickets



After 14 months of treatment

Osteomalacia



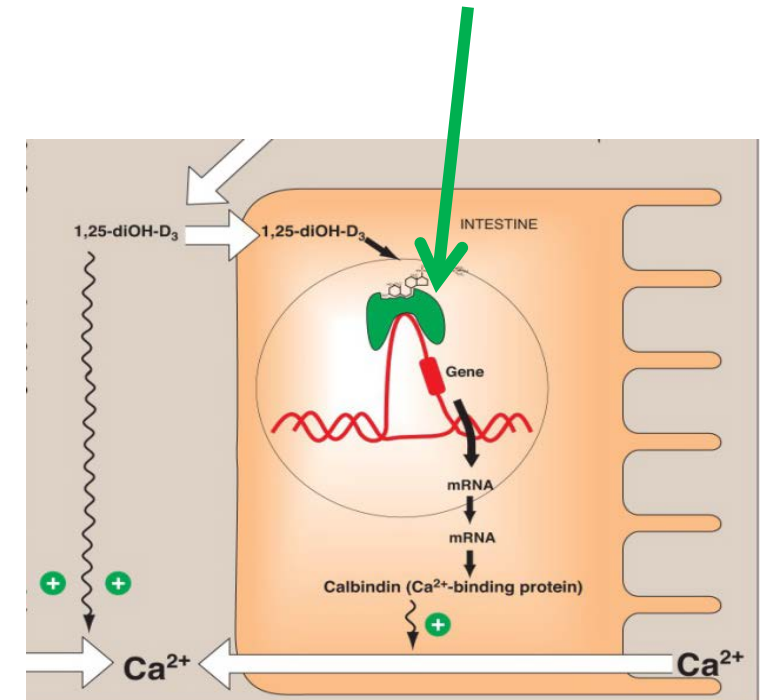
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- Osteomalacia: Vitamin D deficiency in an adult
- Bones are de-mineralized and are susceptible to fracture; May present as non-specific bone pain
- Osteomalacia can be secondary to **reduced exposure to sunlight, dietary deficiency, renal disease or liver disease**
- **Lab findings**
 - **Serum alkaline phosphatase activity is elevated**
 - **Serum calcium and phosphate levels are low**
- Subclinical vitamin D deficiency is common in the population (Teens, young adults and elderly); In teens and young adults vitamin D deficiency affects the attainment of peak bone mass

Mutations in the calcitriol receptor

- Inherited condition where calcitriol is unable to bind to its **receptor** in the intestinal mucosal cells (receptor mutation results in reduced binding of calcitriol)
- As a result, these children have high levels of calcitriol (loss of feedback inhibition)
- **Serum calcium and phosphate levels are low** but **1,25-hydroxyvitamin D levels are high**
- Compare to vitamin D deficient rickets, which has low serum calcium and phosphate and low levels of 25-hydroxyvitamin D

Calcitriol receptor mutation



Hypervitaminosis D

- Hypervitaminosis D is characterized by hypercalcemia (high serum calcium levels) – Due to increased calcium absorption from the diet
- Vitamin D also increases phosphate absorption.
- **Increase in serum calcium and phosphate** results in an increased tendency for **ectopic (soft tissues) mineralization**
- Prolonged hypervitaminosis D (prescription medications) can lead to calcification in soft tissues like kidney.

Summary

1. Formation of cholecalciferol in the skin (action of sunlight)
2. 25-hydroxylation in the liver
3. 1-hydroxylation in the kidney to form 1,25-DHCC
4. 1,25-DHCC binds to its target receptors in intestine to increase synthesis of calbindin
5. 1,25-DHCC acts on bone to increase resorption of bone

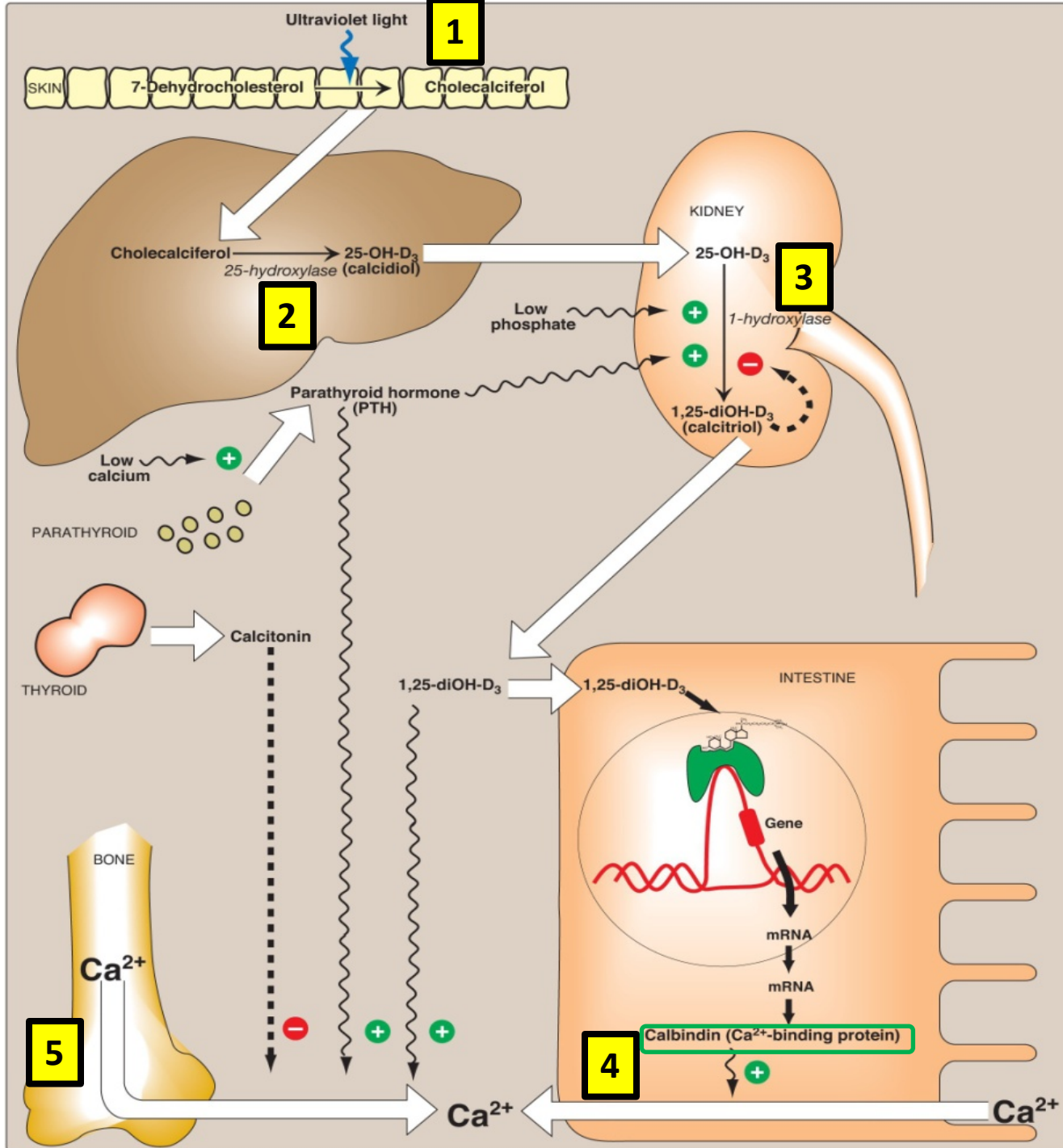


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